

Calibration results

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Camera-system parameters:

cam0 (/camera_0/image_raw):

type: <class 'aslam_cv.libaslam_cv_python.DoubleSphereCameraGeometry'>

distortion: [] +- []

projection: [2.79749048 0.57387295 1347.92612848 1346.79551527 927.26100518

615.34622226] +- [0.00418944 0.0195958 0.22446483 0.22317182 0.11346146 0.14727713]

reprojection error: [-0.000009, 0.000002] +- [0.387327, 0.456949]

cam1 (/camera_1/image_raw):

type: <class 'aslam_cv.libaslam_cv_python.DoubleSphereCameraGeometry'>

distortion: [] +- []

projection: [2.71583935 0.67720491 1314.9277821 1312.77613414 903.10233513

600.03279047] +- [0.00591215 0.02626711 0.32726821 0.32447842 0.22227922 0.18144744]

reprojection error: [0.000000, -0.000001] +- [0.359050, 0.427529]

cam2 (/camera_2/image_raw):

type: <class 'aslam_cv.libaslam_cv_python.DoubleSphereCameraGeometry'>

distortion: [] +- []

projection: [2.95304805 0.40410405 1399.8164263 1397.06821491 981.77162224

579.18403982] +- [0.00361012 0.01904657 0.23329383 0.23121616 0.13840393 0.14456957]

reprojection error: [0.000000, -0.000000] +- [0.267127, 0.301247]

cam3 (/camera_3/image_raw):

type: <class 'aslam_cv.libaslam_cv_python.DoubleSphereCameraGeometry'>

distortion: [] +- []

projection: [2.96808814 0.31866029 1405.75986317 1404.83407467 908.60860824

596.08283871] +- [0.00313564 0.01653073 0.20666023 0.20534791 0.09281142 0.12838488]

reprojection error: [0.000009, -0.000001] +- [0.309157, 0.352924]

baseline T_1_0:

q: [-0.00486261 -0.71799639 -0.00452667 0.69601512] +- [0.00057681 0.00092834 0.00053634]

t: [0.04720252 0.00023143 -0.04994311] +- [0.0002127 0.00009876 0.00027038]

baseline T_2_1:

q: [0.00473787 -0.69760339 -0.00852321 0.71641777] +- [0.00057044 0.00096486 0.00065414]

t: [0.04792472 0.00046123 -0.0486345] +- [0.00032164 0.00009074 0.00020115]

baseline T_3_2:

q: [0.00333819 -0.70331056 -0.00062194 0.71087462] +- [0.00046388 0.00066392 0.00050667]

t: [0.04820172 0.00040643 -0.048481] +- [0.00011673 0.00004812 0.00010838]

Target configuration

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Type: aprilgrid

Tags:

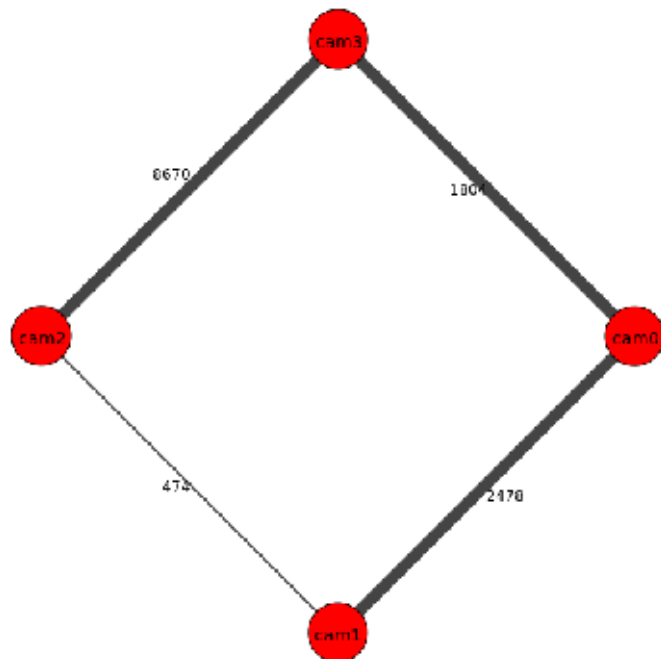
Rows: 6

Cols: 6

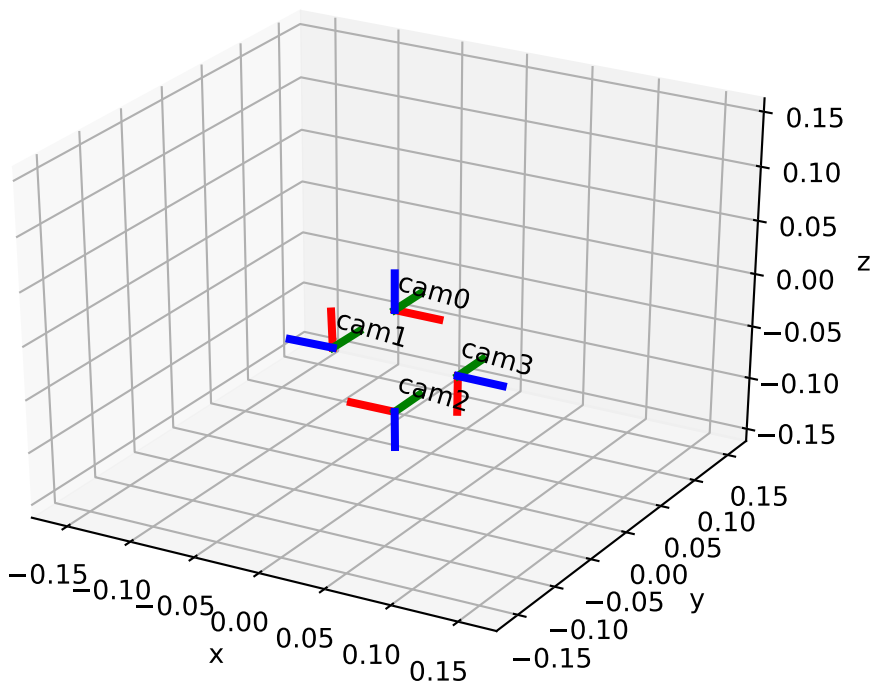
Size: 0.021 [m]

Spacing 0.0059997 [m]

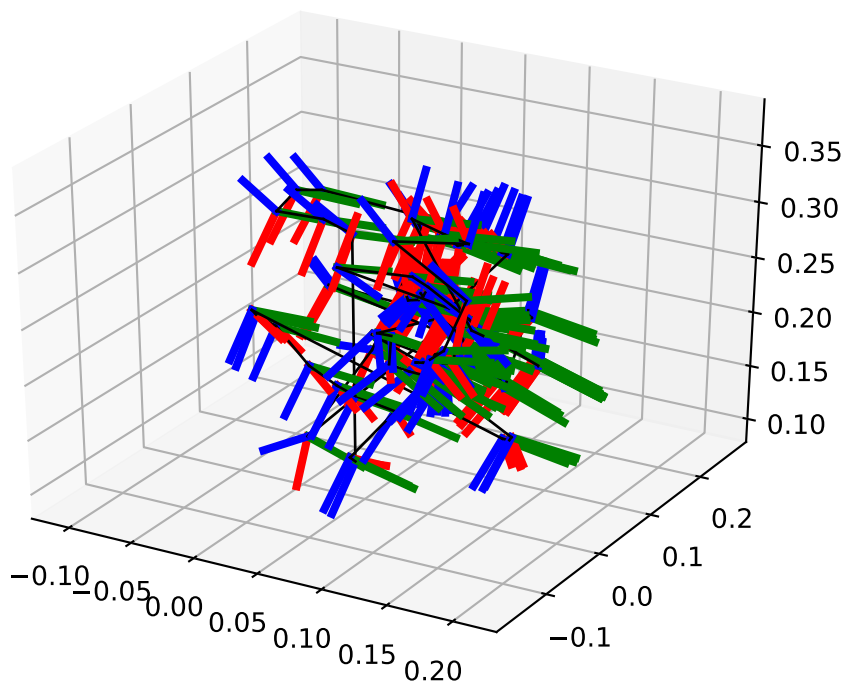
Inter-camera observations graph (edge weight=#mutual obs.)



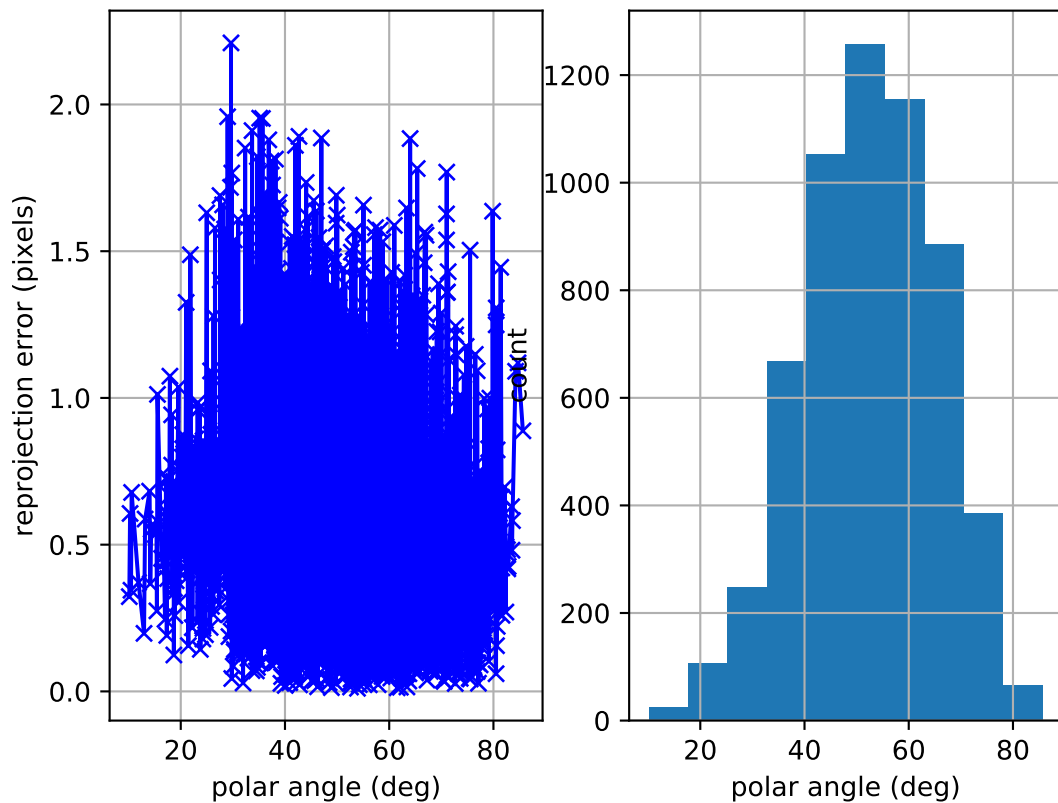
camera system



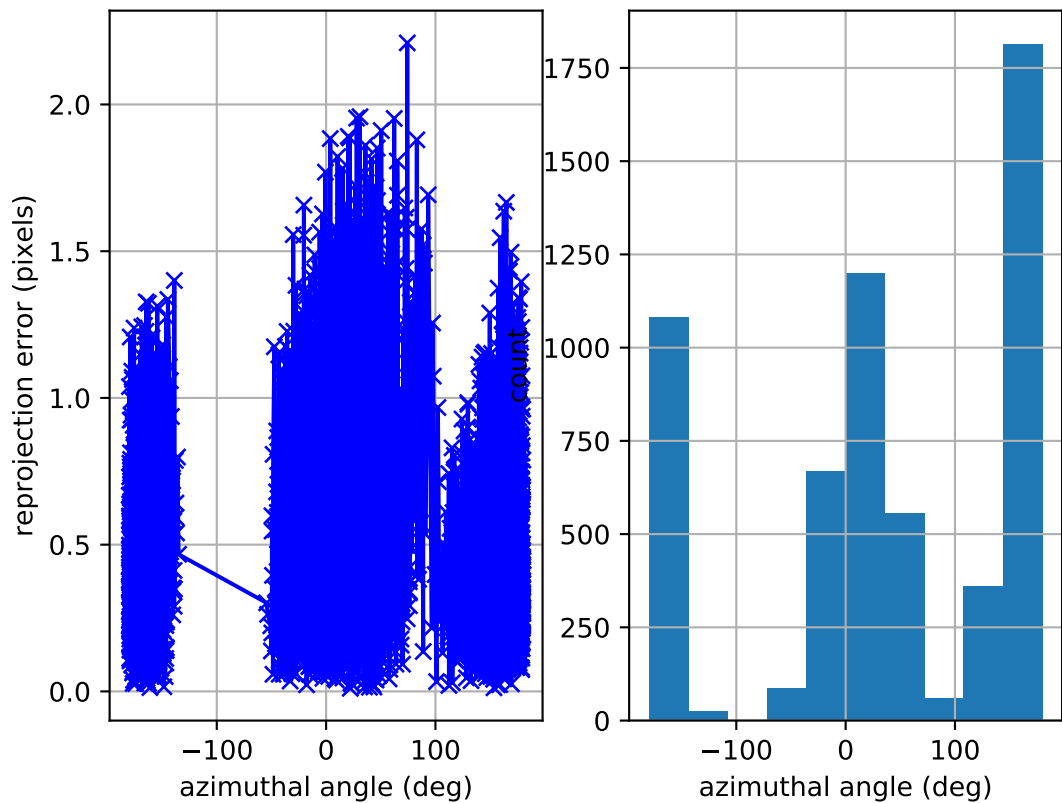
cam0: estimated poses



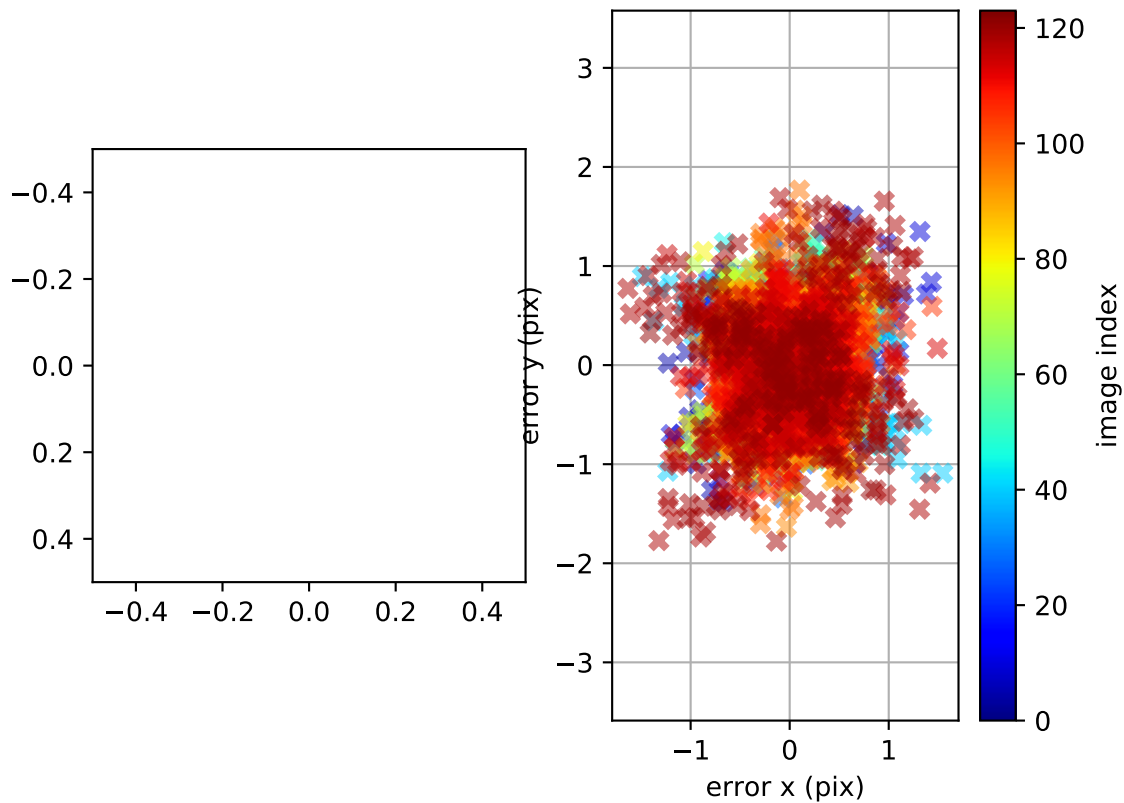
cam0: polar error



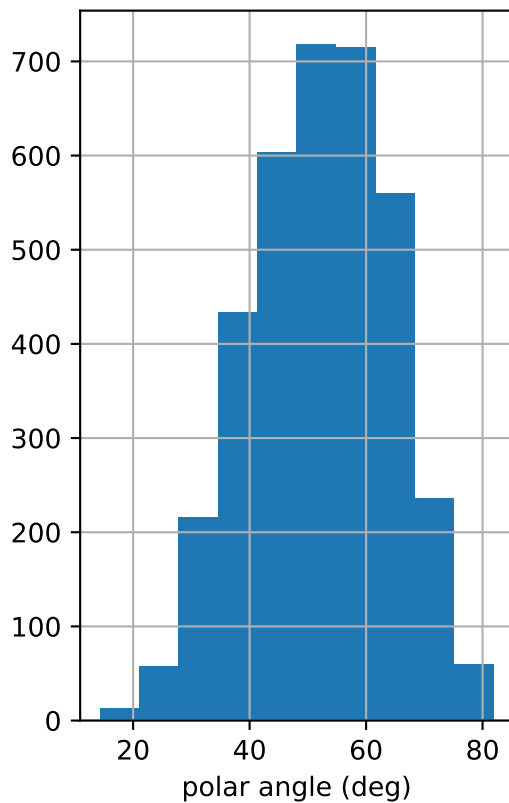
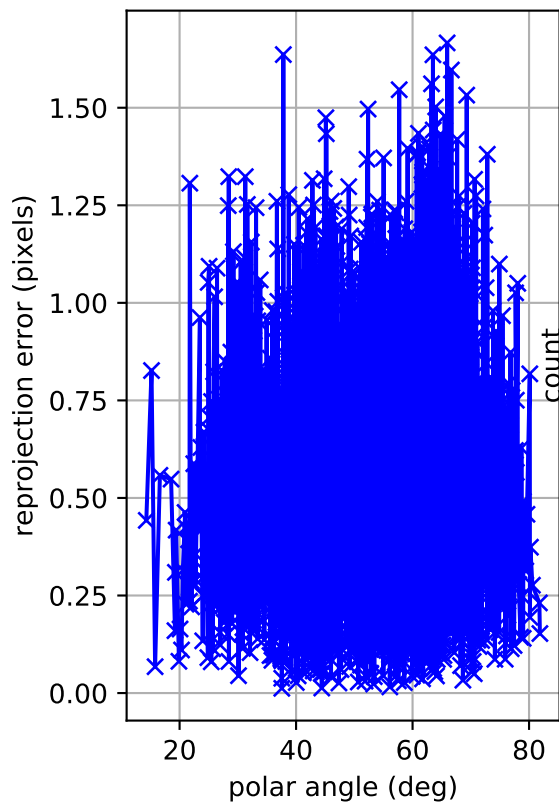
cam0: azimuthal error



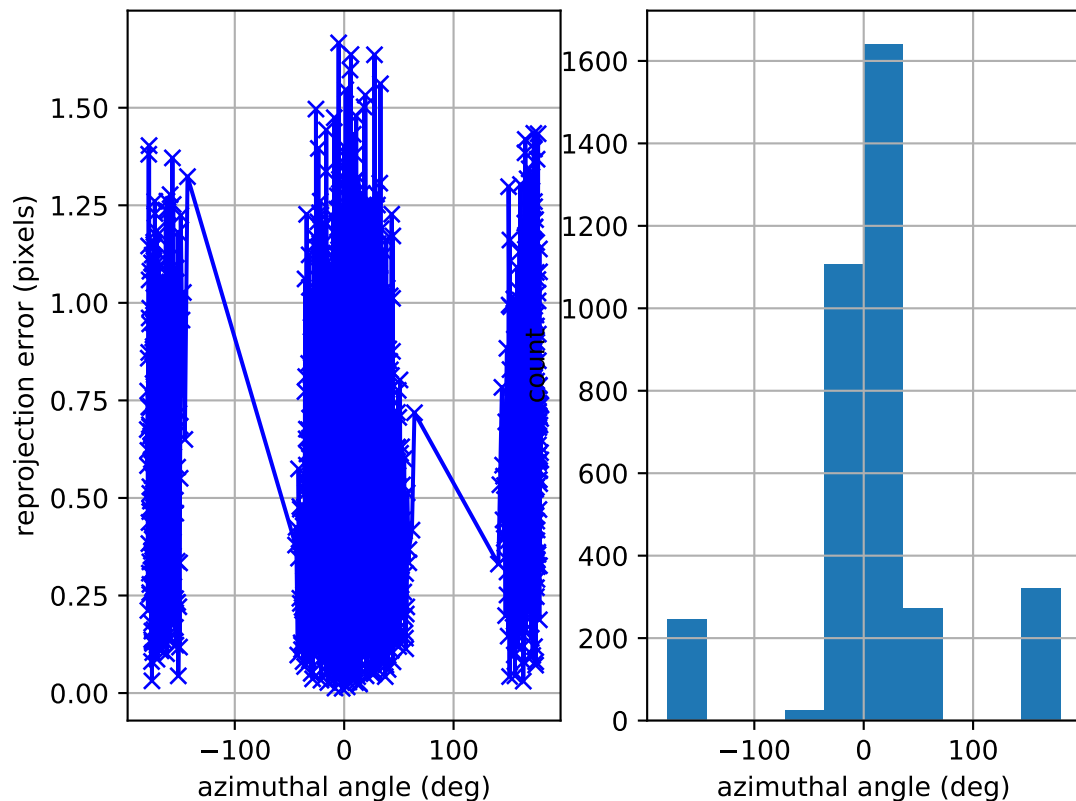
cam0: reprojection errors



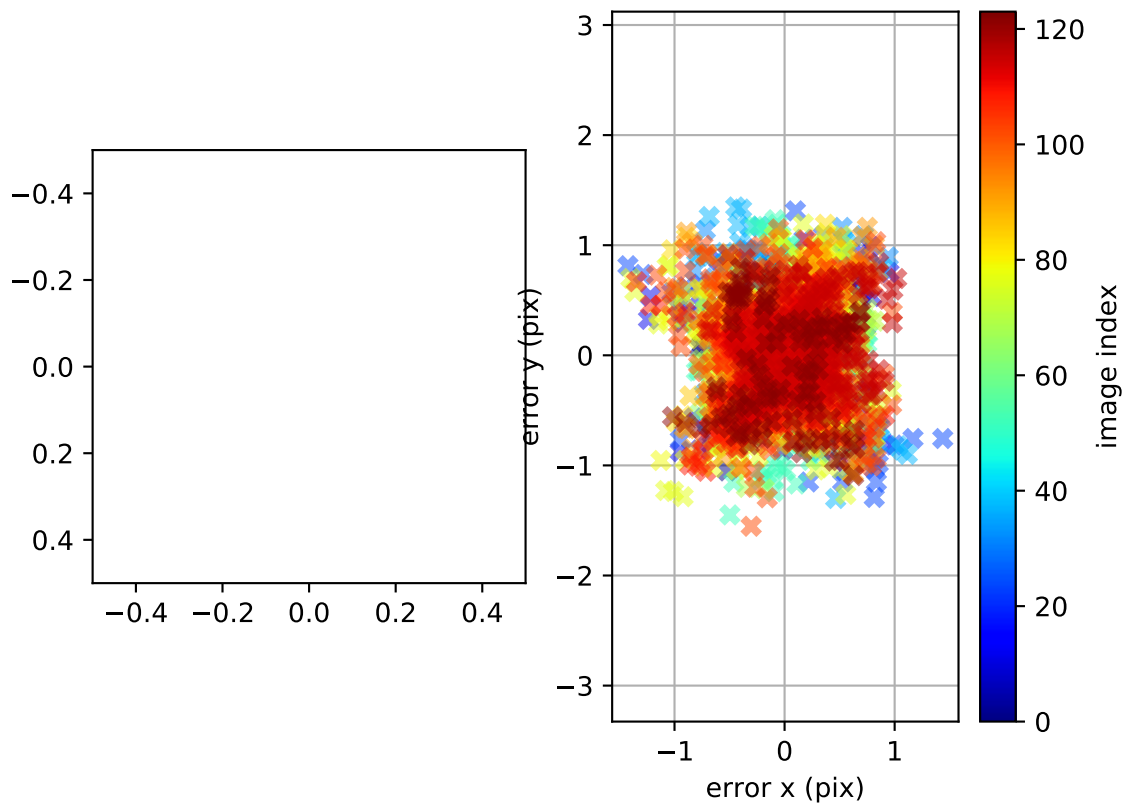
cam1: polar error



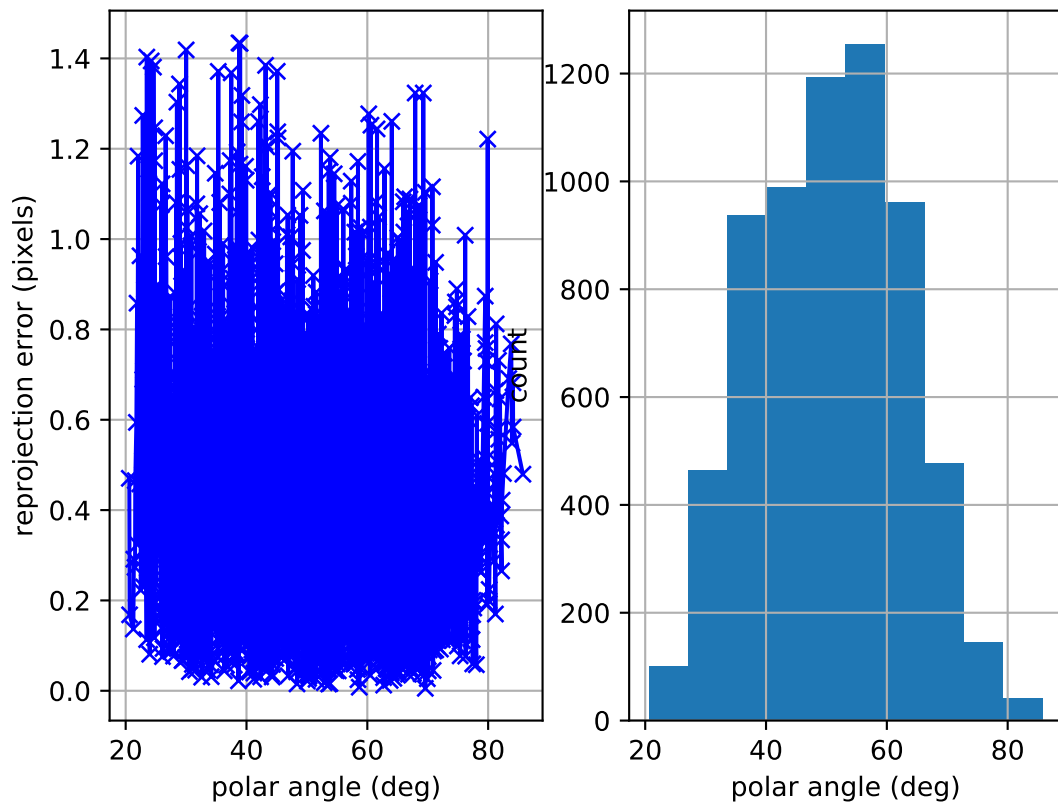
cam1: azimuthal error



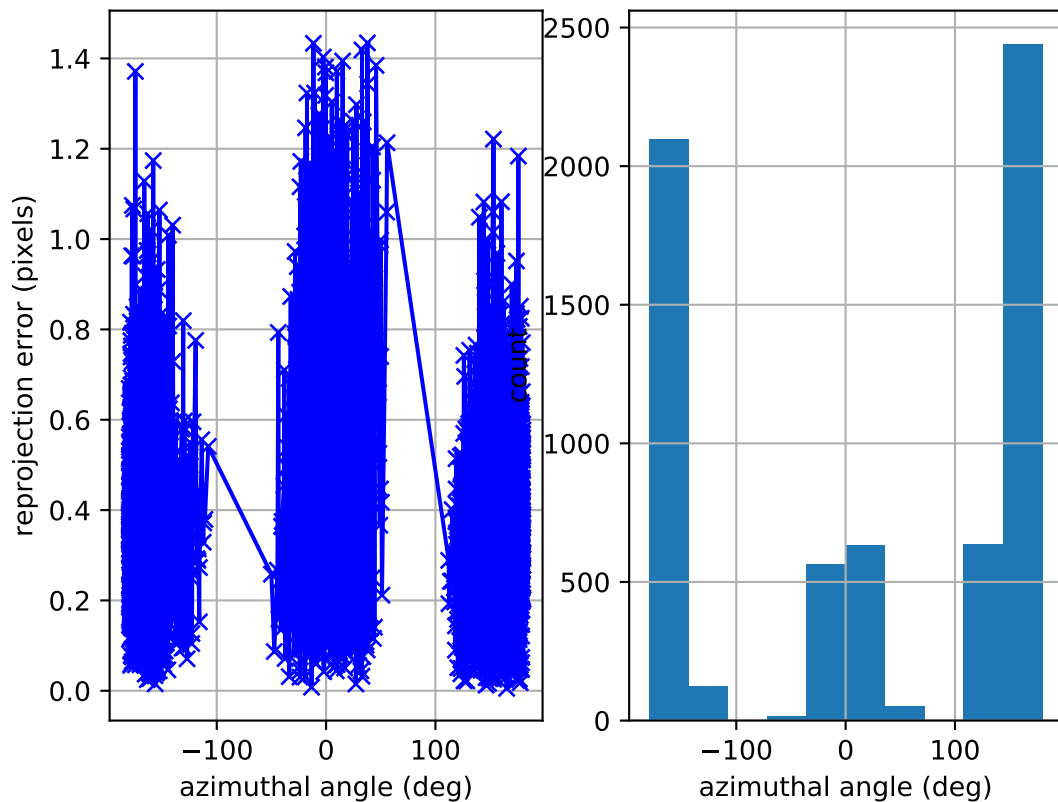
cam1: reprojection errors



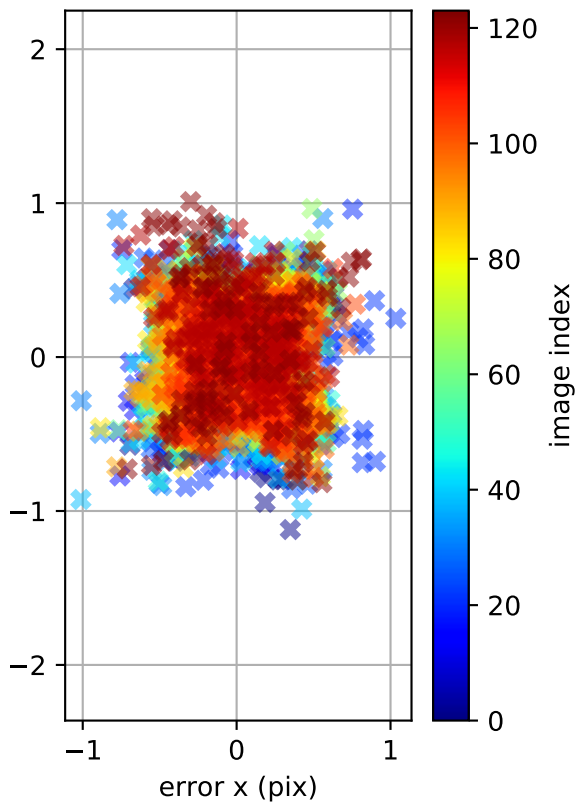
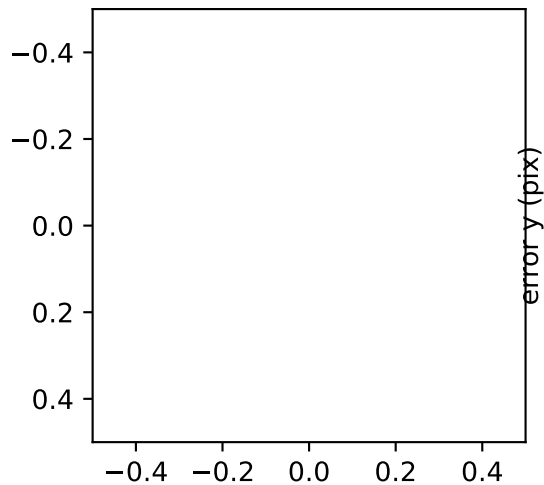
cam2: polar error



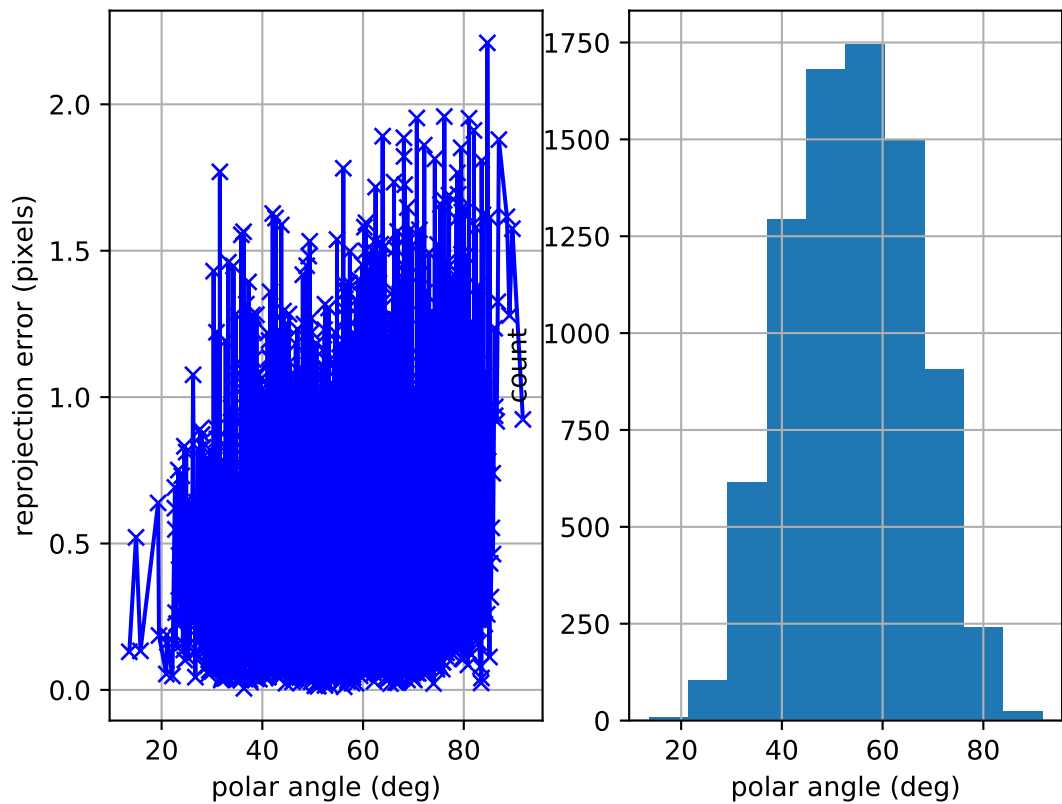
cam2: azimuthal error



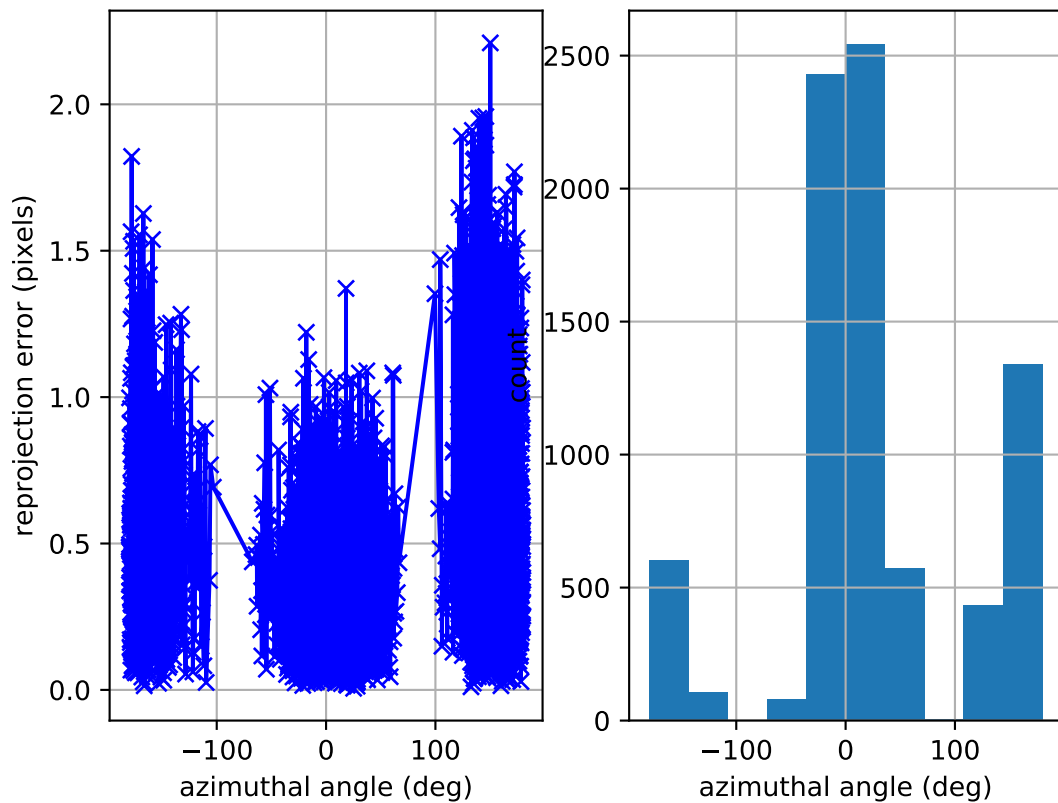
cam2: reprojection errors



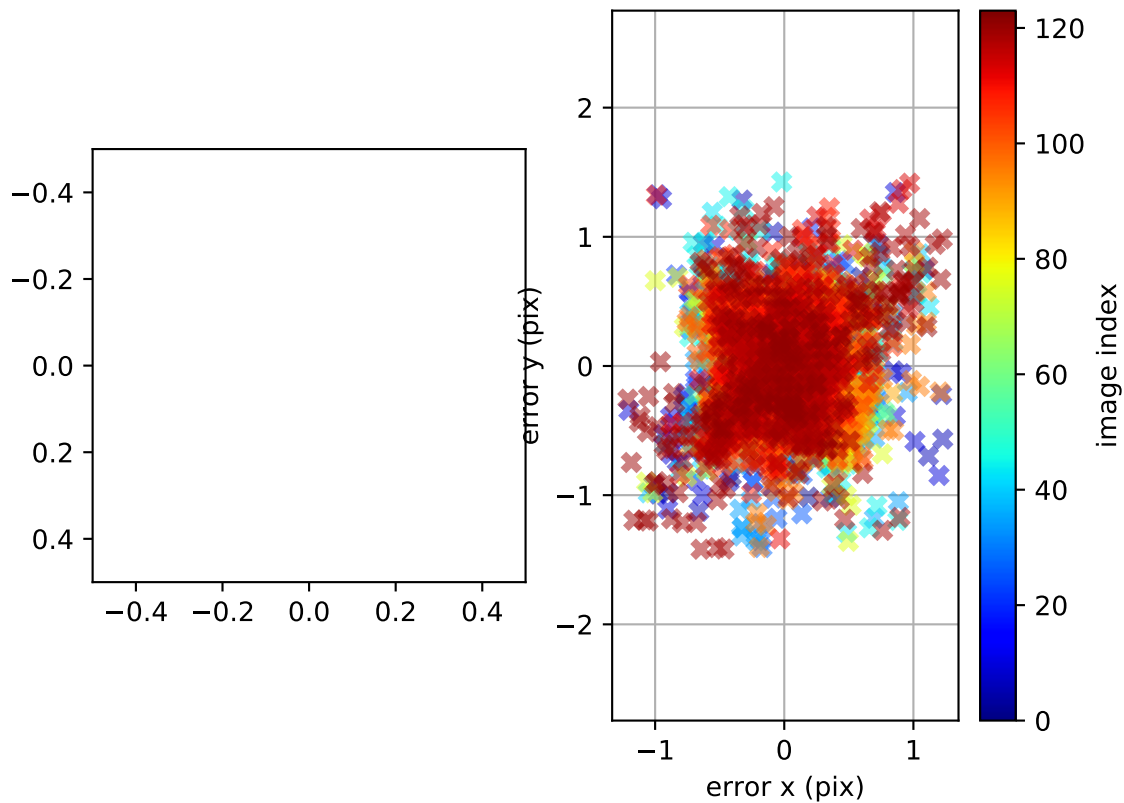
cam3: polar error



cam3: azimuthal error

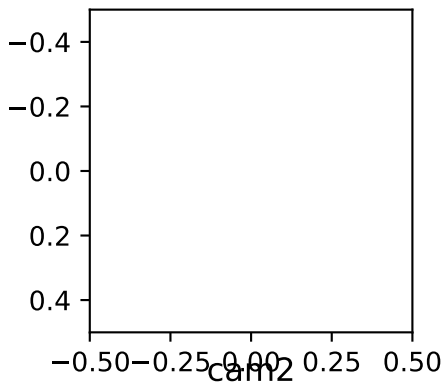


cam3: reprojection errors

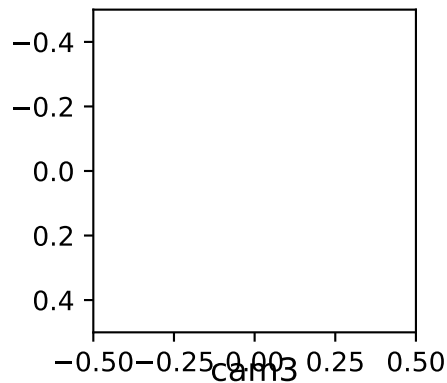


Location of removed outlier corners

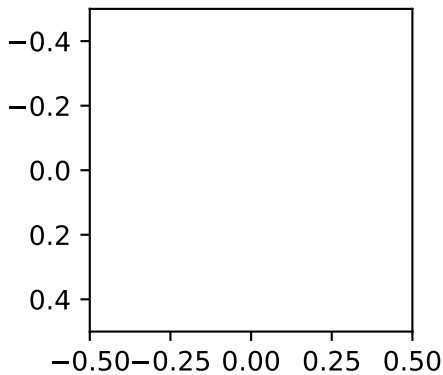
cam0



cam1



cam2



cam3

